

OLYOKMINSK, Russian Federation

WMO: 249440

Lat: 60.400N

Lon: 120.417E

Elev: 753

StdP: 14.30

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-52.1	-48.5	-54.9	0.2	-51.7	-51.2	0.3	-48.5	19.2	0.7	17.0	-2.2	2.9	40	0.698

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	84.7	66.4	81.1	64.9	77.5	63.3	68.9	80.2	66.8	77.4	65.1	74.8	6.0	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.1	95.3	73.2	63.1	89.0	71.4	61.2	83.0	69.7	33.6	80.6	31.9	77.3	30.5	75.4	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.9	14.5	12.6	-54.9	90.2	5.5	2.8	-58.9	92.2	-62.1	93.8	-65.1	95.4	-69.1	97.4
			-54.9	72.1	5.5	1.9	-58.9	73.5	-62.1	74.6	-65.1	75.6	-69.1	77.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	22.0	-23.5	-14.2	5.9	27.4	44.9	60.1	64.9	58.5	42.9	23.1	-4.9	-23.3
	DBStd	33.79	15.80	13.58	13.49	9.48	7.73	7.92	6.70	6.94	7.34	10.79	16.60	16.67
	HDD50	11336	2279	1799	1368	678	197	14	1	12	233	834	1648	2273
	HDD65	15834	2744	2219	1833	1128	623	183	86	220	663	1299	2098	2738
	CDD50	1115	0	0	0	0	39	317	462	277	20	0	0	0
	CDD65	138	0	0	0	0	0	36	82	20	0	0	0	0
	CDH74	1600	0	0	0	0	14	506	862	217	1	0	0	0
	CDH80	421	0	0	0	0	1	125	256	39	0	0	0	0
Wind	WSAvg	5.5	4.8	5.2	5.9	6.6	6.9	5.5	5.1	4.9	5.5	5.8	5.2	4.7
Precipitation	PrecAvg	12.0	0.6	0.4	0.4	0.4	0.9	1.9	2.1	1.8	1.5	0.8	0.8	0.7
	PrecMax	18.4	1.1	0.9	1.2	1.1	2.0	4.1	7.2	3.7	3.5	1.7	1.3	1.5
	PrecMin	7.9	0.2	0.0	0.0	0.1	0.1	0.6	0.4	0.2	0.0	0.2	0.3	0.3
	PrecStd	2.5	0.2	0.2	0.2	0.2	0.6	0.8	1.4	1.0	0.9	0.4	0.2	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.2	23.0	41.7	57.3	75.8	88.1	90.0	84.3	70.4	48.7	34.7	22.2
		MCWB	16.1	21.4	33.5	44.0	55.3	65.5	68.6	66.9	56.4	41.9	31.9	21.5
	2%	DB	10.2	17.0	34.5	50.8	69.8	83.6	86.2	79.7	64.2	43.4	30.6	14.0
		MCWB	9.5	15.7	29.1	39.9	52.5	63.7	67.7	64.8	52.8	38.1	29.0	13.3
	5%	DB	5.1	11.7	29.7	45.9	65.0	79.9	82.6	75.7	59.3	39.9	26.4	9.0
		MCWB	4.5	10.6	25.9	36.8	49.3	62.5	67.1	62.9	50.4	35.6	25.4	8.5
	10%	DB	0.0	6.3	25.7	42.1	60.5	76.1	78.9	71.8	55.2	36.7	19.0	2.3
		MCWB	-0.6	5.4	22.6	34.4	47.2	60.9	65.4	61.0	48.5	33.2	17.8	1.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.3	21.7	33.8	44.6	57.6	68.2	72.2	69.4	57.8	42.7	32.5	21.4
		MCDB	17.4	23.3	41.4	56.4	71.5	81.4	83.5	80.0	66.9	46.8	33.5	22.4
	2%	WB	9.6	15.7	29.4	40.7	53.6	66.0	70.0	66.6	54.3	39.1	29.2	13.4
		MCDB	10.2	16.8	33.9	49.5	66.7	79.4	81.9	76.2	61.3	42.5	30.6	13.9
	5%	WB	4.5	10.6	26.2	37.6	50.8	64.2	68.3	64.6	51.7	36.0	24.9	8.5
		MCDB	5.0	11.8	29.6	44.9	62.5	76.3	79.4	72.9	57.5	39.3	26.4	8.9
	10%	WB	-0.6	5.4	22.6	34.9	48.1	62.4	66.5	62.4	49.2	33.3	17.8	1.9
		MCDB	0.0	6.3	25.5	41.2	58.6	73.6	76.2	69.5	54.4	36.2	19.0	2.4
Mean Daily Temperature Range	5% DB	MDBR	12.3	16.1	21.7	20.2	19.7	22.0	19.8	19.7	16.0	12.3	12.6	11.7
		MCDBR	14.4	17.8	20.8	23.4	28.8	28.9	26.6	26.5	24.5	15.0	14.5	15.9
		MCWBR	14.1	16.7	16.3	14.8	14.6	12.7	11.6	12.6	14.1	10.6	13.9	15.8
	5% WB	MCDBR	14.3	17.7	19.6	21.4	26.1	24.8	23.4	22.6	21.5	13.8	14.2	16.3
		MCWBR	14.0	16.7	15.7	14.0	14.3	11.7	11.2	11.6	13.3	10.6	13.7	16.0
Clear-Sky Solar Irradiance	taub	0.240	0.271	0.299	0.347	0.400	0.416	0.480	0.421	0.334	0.291	0.241	0.211	
	taud	1.979	2.057	2.094	2.138	2.222	2.279	2.087	2.244	2.461	2.375	2.037	1.908	
	Ebn at Noon	156	224	259	267	261	258	235	242	251	224	154	114	
	Edh at Noon	15	25	34	39	39	38	45	36	24	18	13	10	
All-Sky Solar Radiation	RadAvg	153	472	1023	1590	1586	1828	1636	1273	794	383	188	83	
	RadStd	9	23	39	79	137	150	141	95	71	37	22	4	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page